



AADIT AGRAWAL

Computer Science Engineer
AI/ML · HPC · DevOps ·
Security

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LANGUAGES
Python, C/C++, TypeScript,
JavaScript, Rust, Go, Zig,
Bash/Zsh, SQL, Java,
Kotlin, C#, Lua, Swift,
Haskell, Nix, LaTeX, Typst

FRONTEND
Next.js, React, Svelte,
Vue, Astro, Tailwind,
Radix, shadcn/ui

BACKEND
Node.js, Bun, Deno,
Express, tRPC, FastAPI,
Flask, Django

DEVOPS
Docker, Proxmox, Linux,
Git, GitHub Actions,
Ansible, Terraform, Pulumi

NETWORKING / MONITORING
Tailscale, WireGuard,
Nginx, Caddy, Traefik,
HAProxy, Grafana,
Prometheus, Loki

CLOUD
AWS, GCP, Azure, Cloudflare
Workers, Vercel, Fly.io,
Supabase, Convex

AI / ML
PyTorch, JAX, Transformers,
DeepSpeed, FSDP, Ray, W&B,
OpenCV, YOLO, Detectron2,
LangChain, Haystack

LLM / HPC
SGLang, vLLM, llama.cpp,
TensorRT-LLM, MLX, Ollama,
CUDA, RDMA, NCCL

AGENTIC TOOLING
Claude Code, Codex, Pi.dev,
Droid, Cursor, opencode,
ampcode, CodeRabbit

DATABASES
PostgreSQL, MariaDB,
MongoDB, SQLite, Redis,
ChromaDB, pgvector, Milvus

ARCHITECTURE
ARM64, Apple Silicon,
Jetson, AMD64/x86-64, RISC-
V, SIMD

SECURITY
Wireshark, Burp, Nmap,
Metasploit, Ghidra

EDUCATION

Manipal Institute of Technology · BTech – CCE, MAHE

Jul 2023 – May 2027

EXPERIENCE

Head Student Researcher · CASIR, MAHE · ecitis.org

May 2025 – Present

- Architected HPC cluster (96-core Threadripper 7995WX, 48GB A6000, 256GB RAM) – 1,500+ daily inference req, 99.2% uptime.
- Configured a first-of-its-kind-in-India 4x Mac Studio cluster (512GB unified memory, 16TB storage) with Thunderbolt RDMA / InfiniBand-style fabric for MLX/MPs inference and training; leading \$300K NVIDIA Blackwell bring-up.
- Real-time CV on Jetson + DJI drone – 30fps edge inference via RTMP.
- LLM semantic translator for Schneider Electric – IEC 61131-3 → IEC 61499 conversion automation for legacy PLC codebases; migration effort reduced by 70%.
- Technical document ingestion pipeline (Schneider Electric) for machine / electronics engineering support – evolved from RAG + BM25 + keyword cluster indexing to trigram indices with custom pi.dev-based search agents. VLM + LLM stack across 2,000+ docs, 94% accuracy, 3x faster than the prior solution.
- Application security analysis platform (Schneider Electric) – source + compiled-binary evaluation in computer-use sandboxes with OS state monitoring (registry / DLL deltas), pattern-matched against a custom-correlated CVE / CWE / MITRE index. Pentest lead time reduced from 3 weeks → 3 days.
- 3 production AI systems; 2 peer-reviewed publications in progress; team learnings published at blog.ecitis.org.

DevOps Engineer · Pixa AI, Bengaluru

Dec 2025

- Cloud infra + CI/CD for AI apps; 99.5% deployment success across 2 Model API releases.

Web and Design Engineer · [Rossmann New Media 2 Inc.](https://rossmannnewmedia2.com)

Jan 2026 – Present

- Redesigned and redeveloped rossmanngroup.com, migrating it from WordPress to Next.js; moved primary SEO rankings from outside the top 100 into the top 5.

Founding President · [PRISM](https://prism.academy)

Jan 2026 – Present

- Research club (AI/ML, SWE, HPC, DevOps, Biotech, InfoSec) – 150+ members. Academic Advisory Board for AI Curriculum and Adoption.

Management Committee · MIST

Oct 2023 – Jun 2025

- CTF infra for 300+ users, 100% uptime; cut costs by 80%.
- 12+ workshops for 150+ participants; top 10% national CTFs.

RESEARCH

FL + Digital Twin for Route Optimisation · SCE, MAHE

Jun 2024 – Jul 2025

- GPS telemetry (500K+ datapoints); dashcam CV via OpenCV + YOLO – 12,000+ labelled samples.

PROJECTS

Homelab Infrastructure · Self-hosted cloud

2022 – Present

- 70+ services on Proxmox/Docker – 98% uptime, 4 yrs, 8TB MergerFS, Blackwell-GPU, 3-2-1 backups.
- Fine-tuned models for security (30fps) + 4K transcoding; 95% CPU to GPU offload.
- Mitigated live DDoS (50K+ req/min); 95% attack surface reduction.

The TIK Menu API · [GitHub](https://github.com)

2024 – 2025

- 5,000+ DAU, 150K+ monthly req, campus adoption in under 7 days. REST API + Cloudflare R2.

Coolstuff.work Forgejo

2023 – Present

- Suite of utility tools solving everyday problems for peers – 10,000+ aggregate DAU across sub-apps.
- OR-Free: Free AI model tracker on OpenRouter; real-time availability and pricing dashboard.
- Contact Cards + Mass Contact Maker: vCard QR generator + bulk VCF creation from Excel sheets.

- › [Campus Directory](#): Crowdsourced campus contact hub with real-time status and fuzzy search.
- › [MIT Courses Search](#): Course discovery with interactive physics graph (500+ nodes), fuzzy search. Next.js 15, Radix UI, Fuse.js.
- › [transcript-cli](#): Multi-runtime video transcription CLI tool.
- › Sindhi Menu: Cyclic mess menu viewer.

[ophoner](#) · *Mobile agent control loop*

2026 - Present

- › Phone-based agent loop for device APIs and filesystem workflows – extensible via tmux and Android shell tooling. Focused on on-the-go file editing/querying without heavy desktop dependencies.

Developer Tooling (OSS)

2025 - Present

- › [agentskills](#): Published library of Claude Code skills for reusable agentic development workflows.
- › [t3code](#) (fork): customised for parallelised agent-driven development flows.
- › Self-hosted CI + sandboxing stack (migrating from Daytona); CodeRabbit integrated for public-repo review.